Requirements specification for Rent business process

1. General description of business process

The customer enters the rental point and chooses a car and the rental period. The employee, logged in to the rental system, checks the availability of the car, verifies the customer's identity document and driving licence. A new customer fills in a form with personal data, which the employee then enters into the client EXCEL sheet (an existing client is identified by PESEL number). The employee assigns the chosen car (identified by its licence plate) to the rental. The system generates a rental agreement with the date of issue, the planned return date and the cost, calculated from the daily price of the car and the number of days. The customer pays the cost. When the car is returned, the employee records the actual return date; if the car is returned later than planned, the cost is recalculated for the actual number of days. Metrics: car utilization rate, number of idle days, number of rentals, number of rental days, average rental length and rental revenue. Analytical problems: explaining month-to-month changes of utilization and finding the periods, rental points, makes and fuel types for which cars stay idle.

The business goal of the company is to increase the car occupation time, i.e. to make sure that cars do not stay unrented. The goal is measured by the utilization rate of cars in individual rental points (the number of days cars were rented divided by the number of days they were available), which should rise, and by the number of idle days (days when an available car was not rented) within a month, which should fall. The analytical problems below concern the factors that influence these two metrics.

2. Data sources structures

RentMaster

TABLE NAME	ATTRIBUTE	ATTRIBUTE TYPE	DESCRIPTION
Rent	A rental of a car by a client, handled by an employee of a rental point. Identified by rent_id.		
	rent_id	INTEGER	PK, auto generated starting from 1 with increment 1
	client_PESEL	VARCHAR(11)	client's PESEL

			number
	car_licence_plate	VARCHAR(255)	FK, licence plate of the rented car (table Car)
	employee_id	INTEGER	FK, id of the employee who handled the rental (table Employee)
	date_issued	DATETIME	date of rent issuance, DD-MM-YYYY; HH:mm
	date_returned_planned	DATETIME	date of the expected rental return, DD-MM-YYYY; HH:mm
	date_returned_actual	DATETIME	date of the actual return of the rental, DD-MM-YYYY; HH:mm
	cost	DECIMAL	total cost of the rental in PLN
	mileage_before	INTEGER	mileage of the rented car before rent
	mileage_after	INTEGER	mileage of the rented car after rent
Employee	An employee of a rental point who handles the rentals. Identified by employee_id.		
	employee_id	INTEGER	PK, auto generated starting from 1 with increment 1

	Name	VARCHAR(255)	first name
	Surname	VARCHAR(255)	last name
	location_id	INTEGER	location id
Location	A rental point of the car rental network, located in one of the city districts. Identified by location_id.		
	location_id	INTEGER	PK, auto generated starting from 1 with increment 1
	City	VARCHAR(255)	city of the location
	Street	VARCHAR(255)	street of the location
	Name	VARCHAR(255)	name of the location
Car	A car of the fleet, identified by its licence plate.		
	licence_plate	VARCHAR(255)	PK, licence plate of the car
	location_id	INTEGER	location id
	Brand	VARCHAR(255)	brand of the car
	Model	VARCHAR(255)	model of the car
	Color	VARCHAR(255)	color of the car

	Mileage_current	INTEGER	current mileage of the car in km
	Price_per_day	DECIMAL	price of renting the car for a day in PLN
	car_year	YEAR	year of the car production, YYYY
	fuel_type	VARCHAR(255)	fuel type of the car; allowed values: petrol, diesel, LPG, electric, hybrid
	type	VARCHAR(255)	type of the car; allowed values: suv, van, combi, sedan, coupe, sport
Service	A single service of a car in a workshop (routine, accident or malfunction). Identified by service_id.		
	service_id	INTEGER	PK, auto generated starting from 1 with increment 1
	car_licence_plate	VARCHAR(255)	FK, licence plate of the serviced car (table Car)
	start_date	DATE	date of car's arrival in the service; DD-MM-YYYY
	end_date	DATE	date of car's return from the service; DD-MM-YYYY
	mileage	INTEGER	mileage of the car at the moment of service in km

	cost	**DECIMAL**	**total cost of the service in PLN**
	type	**VARCHAR(255)**	**type of the service; allowed values: routine, accident, malfunction**
	description	**VARCHAR(255)**	**optional description of the service**

CEO Excel

Sheet 1 (information about clients renting cars, manually entered from a paper form filled by clients, line 1 is a header row):

Column A - Client's first name (text),
Column B - Client's last name (text),
Column C - Client's PESEL number (text),
Column D - Client's age (derived from PESEL) (numeric),
Column E - Client's gender (derived from PESEL) (text),
Column F - Client's driver's licence number (text),
Column G - Client's phone number (text).

3. Scenarios of analytical problems

1. Why was there an increase / decrease of idle time of cars this month?
    1. Compare the idle time of rentals of different fuel types in the analyzed month relative to previous months.
    2. What car brands and models have the lowest idle time in the current and the previous month?
    3. Compare individual employees in terms of the utilization rate of rentals handled in the current and the previous month.
    4. What age group is responsible for the majority of utilization rate?
    5. Does utilization rate increase / decrease in specific periods?
    6. What is the client's satisfaction with a rented car?
    7. Which gender tends to choose which fuel type?

2. What is the influence of the car's usage on the utilization rate and idle time of the car?
    1. What is the utilization rate of the cars in relation to the mileage at the beginning of the month?
    2. What is the average mileage at service for individual fuel types, brands and car age categories?
    3. Which gender drives more km per rental day?
    4. What is the car's idle time in relation to cost of servicing?

5. What is the utilization rate of cars exceeding a certain mileage/age, by client age group?

4. Data needed for analytical problems

Analytical problem “Why was there an increase / decrease of idle time of cars this month?”

1. Compare the idle time of rentals of different fuel types in the analyzed month relative to previous months.
    - **idle time** (calculated as days of the month - days in service - days rented) - RentMaster, table Rent, columns date_issued and date_returned_actual, and table Service, columns start_date and end_date, grouped by car_licence_plate
    - **fuel type** - RentMaster, table Car, column fuel_type
    - **month of rental** - RentMaster, table Rent, column date_issued

2. What car brands and models have the lowest idle time in the current and the previous month?
    - **idle time** (calculated as days of the month - days in service - days rented) - RentMaster, table Rent, columns date_issued, date_returned_actual, and car_licence_plate, and table Service, columns start_date and end_date
    - **car brand and model** - RentMaster, table Car, columns Brand and Model
    - **month of rental** - RentMaster, table Rent, column date_issued

3. Compare individual employees in terms of the utilization rate of rentals handled in the current and the previous month.
    - **utilization rate** - RentMaster, table Rent, columns date_issued and date_returned_actual, and table Service, columns start_date and end_date (utilization rate = rental days / available days, where available days = days of the month - days in service)
    - **employee name** - RentMaster, table Employee, columns Name and Surname

4. What age group is responsible for the majority of utilization rate?
    - **utilization rate** - RentMaster, table Rent, columns date_issued and date_returned_actual, and table Service, columns start_date and end_date (utilization rate = rental days / available days, where available days = days of the month - days in service)
    - **client age group** - CEO Excel, Sheet 1, column D (Client's age), we define age groups as following: 18-24, 25-34, 35-44, 45-54, 55-64, 65+

5. Does utilization rate increase / decrease in specific periods?
    - **utilization rate** - RentMaster, table Rent, columns date_issued and date_returned_actual, and table Service, columns start_date and end_date (utilization rate = rental days / available days, where available days = days of the month - days in service)

- **specific periods** must be collected from a publicly available calendar, e.g., Google Calendar

6. What is the client's satisfaction with a rented car?
   - **rental details** - RentMaster, table Rent, check for entity existence
   - **client's satisfaction** - no such information

7. Which gender tends to choose which fuel type?
   - **client gender** - CEO Excel, Sheet 1, column E (Client's gender)
   - **fuel type** - RentMaster, table Car, column fuel_type, joined via table Rent

Analytical problem "What is the influence of the car's usage on the utilization rate and idle time of the car?"

1. What is the utilization rate of the cars in relation to the mileage at the beginning of the month?
   - **utilization rate** - RentMaster, table Rent, columns date_issued and date_returned_actual, and table Service, columns start_date and end_date (utilization rate = rental days / available days, where available days = days of the month - days in service)
   - **mileage at the beginning of the month** - RentMaster, table Rent, column mileage_before (taking the value from the first recorded rental of the given month for a specific car_licence_plate)

2. What is the average mileage at service for individual fuel types, brands and car age categories?
   - **mileage at service** - RentMaster, table Service, column mileage
   - fuel type, brand and car age **- RentMaster, table Car, columns fuel_type, Brand and car_year; age categories (up to 1 year, 1 to 3 years, 3 to 5 years, over 5 years) are calculated from car_year and the service date (table Service, column start_date)**

3. Which gender drives more km per rental day?
   - **kilometers driven** - RentMaster, table Rent, calculated as the difference between mileage_after and mileage_before
   - **rental days** - RentMaster, table Rent, calculated as the difference between date_returned_actual and date_issued
   - **client gender** - CEO Excel, Sheet 1, column E (Client's gender)

4. What is the car's idle time in relation to cost of servicing?
   - **idle time** (calculated as days of the month - days in service - days rented) - RentMaster, table Rent, columns date_issued and date_returned_actual, and table Service, columns start_date and end_date, grouped by car_licence_plate
   - **cost of servicing** - RentMaster, table Service, column cost

5. What is the utilization rate of cars exceeding a certain mileage/age, by client age group?

- **utilization rate** - RentMaster, table Rent, columns date_issued and date_returned_actual, and table Service, columns start_date and end_date (utilization rate = rental days / available days, where available days = days of the month - days in service)
- **car mileage/age threshold** - RentMaster, table Rent, column mileage_before, and table Car, column car_year
- **client age group** - CEO Excel, Sheet 1, column D (Client's age), we define age groups as following: 18-24, 25-34, 35-44, 45-54, 55-64, 65+

It is not possible to build a BI system to answer analytical question 1.6 ("What is the client's satisfaction with a rented car?") without introducing additional activities in the rental process.

We suggest introducing an automated post-rental phone message survey system triggered 3 hours after a car is returned. Such questionnaire should contain as minimum the following set of questions:

- How would you rate the overall condition of the car? (Numeric from 1 to 5)
- Did you experience any mechanical issues during your rental? (Available values: Yes, No)
- How likely are you to rent this specific vehicle model again? (Numeric from 1 to 5)

Survey results would be automatically uploaded to an EXCEL sheet. Sample structure of the survey sheet (each row reflects a single rental):

- Column A - rent_id (FK pointing to the specific rental)
- Column B - car_condition_rating (integer, 1-5)
- Column C - experienced_issues (boolean)
- Column D - likelihood_to_rent_again (integer, 1-5)
- Column E - additional_comments (text)

